Functional Requirements Document – Employee Management

TaskFlow – Intelligent Task Allocation & Tracking System

> **Document:** Functional Requirements Document (sub-module EMP) · **Version:** 1.0 · **Status:** Draft **Project type:** Personal Project (Skill Development) · **Date:** 31-Oct-2026 **Author:** Jojo George (jojogeorge3344@gmail.com) **Module:** TaskFlow Management · **Sub-module:** 03.1 Employee Management (code `EMP`) **Based on:** `BRD-Employee-Management.md` v1.0 © 2026 Jojo George. All rights reserved.

Document Control

Revision History

Version	Date	Author	Summary of changes
1.0	31-Oct-2026	Jojo George	Initial version, written from the Employee Management BRD v1.0 and TaskFlow as built

Related Documents

Document	Purpose	Location
BRD – Employee Management	Business needs this FRD details (`REQ-EMP-nn` , `BR-EMP-nn`)	`../../../BRD/03-TaskFlow-Management/03.1-Employee-Management/BRD-Employee-Management.md`
BRD – Overall	Roles, global rules `BR-nn` , global NFRs	`../../../BRD/00-Overall/BRD-TaskFlow-Overall.md`
FRD – Overall	Shared roles, Authentication, cross-cutting behaviour, index	`../../00-Overall/FRD-TaskFlow-Overall.md`
Related FRDs	Project Management, Department Management	`../03.2-Project-Management/FRD-Project-Management.md` , `../03.3-Department-Management/FRD-Department-Management.md`
Development Plan	Screen 5 – Employee Management	`TF - Documents/Screens Wise Feature Development Plan/05-employee-management-screen.md`
HLD / LLD	Technical design	`../../../HLD/03-TaskFlow-Management/03.1-Employee-Management/HLD-Employee-Management.md` , `../../../LLD/03-TaskFlow-Management/03.1-Employee-Management/LLD-Employee-Management.md`
TDD	Technical design document (whole system)	`TF - Documents/TF - TDD/TDD-TaskFlow.md`

1. Introduction

1.1 Purpose

This FRD specifies the detailed, testable behaviour of the Employee Management sub-module: the employee list, the employee details side panel and the add / edit employee side panel – their inputs, outputs, rules, validations, messages and permissions.

1.2 Scope

In scope: listing, searching and filtering employees; viewing one employee's details with derived managers; adding an employee with role, departments, project allocations, temporary password and status; editing code, name, username, email and role; activating and deactivating; soft-deleting with safety checks; resetting an employee's authenticator MFA.

Out of scope (BRD section 3.2): changing departments or allocations after creation (Department Management and Project Management), setting another person's password, self-registration, bulk import / export, a Department filter on the list.

1.3 Audience

Developer, tester, reviewer and product owner of TaskFlow (all Jojo George).

1.4 Conventions

- Requirement IDs: `FR-EMP-nn`. Use cases: `UC-EMP-nn`. Business rules: `BR-EMP-nn` (Employee Management BRD) and `BR-nn` (overall BRD). BRD requirements: `REQ-EMP-nn`.
- Priority: **Must / Should / Could** (MoSCoW), taken from the BRD requirement traced.
- "The system shall …" = mandatory behaviour.
- `{name}` / `{Full name}` / `{n}` in messages are replaced by the real value.
- Permission "System" = a rule the system applies whatever the user's permissions.

1.5 Definitions & Abbreviations

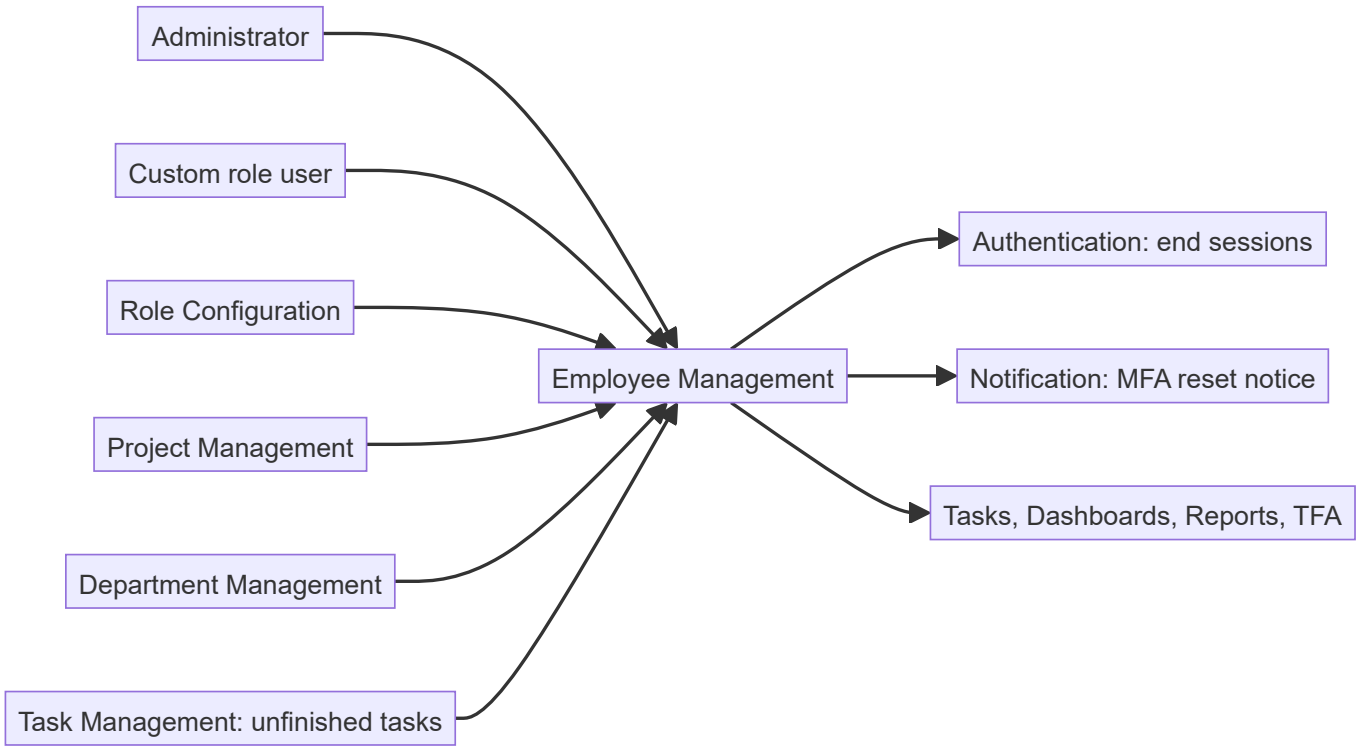
Term	Meaning
Employee	Any person with a TaskFlow sign-in account, whatever their role
Live employee	An employee who has not been deleted
Allocation	The whole percentage (1–100) of a person's time given to one project, with a position
Position	A label for a person's job on a project (for example Developer); it grants no access
Derived manager	A manager worked out from a person's departments and projects, never typed in
Soft delete	Hiding a record from every list while keeping it and its history
MFA	Two-step verification with an authenticator app
Session	A signed-in state on a device; ending it forces a new sign-in
Employee action permission	Any of `employees.create`, `employees.edit`, `employees.status.change`, `employees.delete`, `employees.role.assign`, `employees.mfa.reset`
Scope	Own / Team / Organisation data visibility; Employee Management is always Organisation

2. System Overview

2.1 Module Map

Code	Module	Child modules / screens	Primary roles
EMP	TaskFlow Management → Employee Management (this FRD)	Employee Management list, Employee details (side panel), Add / Edit employee (side panel)	Administrator, custom roles with employee action permissions
PROJ	Project Management (related)	Supplies projects and positions; owns allocations after creation	Administrator, Manager
DEPT	Department Management (related)	Supplies departments; owns memberships after creation	Administrator
AUTH	Authentication (overall FRD)	Sessions ended by employee actions; MFA	All users
NOTIF	Notification (related)	Delivers the MFA-reset notification	All users
ROLE	Role Configuration (related)	Supplies the roles offered in the form and filter	Administrator

2.2 Context Diagram



3. User Roles & Permissions

3.1 Roles

Role	Description	Data scope
Administrator	Full system owner; adds, edits, activates / deactivates, deletes employees, assigns any role, resets MFA	Organisation
Manager	Holds `employees.view` only; no menu item or screen here, but other screens (for example the task assignee picker) list people with it	Organisation list, read only, through other screens
Employee	No access to this sub-module	None
Custom role	Sees the screen only with at least one employee action permission; can do what it is granted	As granted (Organisation)

3.2 Permission Matrix (functional view)

Module › Function	Permission key	Administrator	Manager	Employee	Custom role
Employees › View list and details	`employees.view`	Granted	Granted	Denied	Configurable
Employees › Add employee	`employees.create`	Granted	Denied	Denied	Configurable
Employees › Edit employee	`employees.edit`	Granted	Denied	Denied	Configurable
Employees › Activate / deactivate	`employees.status.change`	Granted	Denied	Denied	Configurable
Employees › Choose a role other than Employee	`employees.role.assign`	Granted	Denied	Denied	Configurable
Employees › Reset two-step verification	`employees.mfa.reset`	Granted	Denied	Denied	Configurable
Employees › Delete employee	`employees.delete`	Granted	Denied	Denied	Configurable
Settings › Give or take away the Administrator role (used here)	`roles.administrator.assign`	Granted	Denied	Denied	Denied (not assignable)
Projects › Fill the Project filter and allocation lists (supporting)	`projects.view`	Granted	Granted	Denied	Configurable
Departments › Fill the Departments list (supporting)	`departments.view`	Granted	Granted	Denied	Configurable

Every `employees.*` key except `employees.view` also requires `employees.view`. Older roles may still hold `employees.manage`, which acts like the finer employee keys it was split into.

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4. Functional Requirements by Module

4.1 Module: Employee Management (EMP)

Purpose: Keep employee sign-in accounts accurate and control who can sign in. · **BRD reference:** BRD-Employee-Management SCREEN 1–3, BR-EMP-01 – BR-EMP-14; overall BR-02, BR-21, BR-24, BR-25, BR-26

4.1.1 Screen: Employee Management (list)

Field	Value
Route / location	`/employees` – menu **TaskFlow Management → Employee Management**
Access	Administrator; custom roles holding at least one employee action permission (list data needs `employees.view`). Manager and Employee: no access (menu hidden, route redirects to their dashboard)
Entry points	Side menu; **View employees** on a Project Management card (opens the list filtered by that project)

Change from original BRD (Screen 5): the Department filter is replaced by Project, Role and Status filters (departments are a column); username and projects are added to the columns; deletion is a soft delete blocked only by **unfinished** assigned tasks; own-account and last-Administrator protection are new.

Screen elements

Element	Type	Description	Mandatory	Visible to
Employee count	Header text	"{n} employees", plus " match the current filters" when a filter is set	–	All with screen access
Add employee	Button	Opens the Add employee side panel	–	Users with `employees.create`
Search	Text input	Placeholder "Name or email"; also matches username	No	All with screen access
Project	Drop-down	Filter by project (all projects)	No	All with screen access
Role	Drop-down	Employee, Manager, Administrator, plus custom roles	No	Custom roles listed only to users with `employees.role.assign`
Status	Drop-down	All statuses / Active / Inactive	No	All with screen access
Reset	Button	Clears every filter	–	All with screen access
Employee	Column	Full name (link to details) and username	–	All with screen access
Email	Column	Email address	–	All with screen access
Projects	Column	Each project with percentage and position; a total when more than one	–	All with screen access
Departments	Column	Department names	–	All with screen access
Role / Status	Columns	Role name; Active or Inactive	–	All with screen access
Edit	Row icon (pencil)	Opens the Edit employee side panel	–	Users with `employees.edit`
Activate / Deactivate	Row icon (power)	Switches the account on or off	–	Users with `employees.status.change`
Delete	Row icon (trash)	Soft-deletes the employee	–	Users with `employees.delete`; never on the user's own row

Functional requirements

ID	Requirement	Priority	Permission	BRD trace
FR-EMP-01	The system shall list every live (not deleted) employee, ordered by full name.	Must	`employees.view`	REQ-EMP-01, BR-EMP-07
FR-EMP-02	The system shall show per row the employee's full name and username, email, projects (each with percentage and position), departments, role and status.	Must	`employees.view`	REQ-EMP-02
FR-EMP-03	The system shall show a project total in the Projects column when an employee has more than one project.	Should	`employees.view`	REQ-EMP-03
FR-EMP-04	The system shall show the number of employees listed as "{n} employees" and append " match the current filters" while any filter is set.	Should	`employees.view`	REQ-EMP-04
FR-EMP-05	The system shall filter the list to employees whose full name, email or username contains the search text, ignoring case.	Must	`employees.view`	REQ-EMP-05
FR-EMP-06	The system shall filter the list to employees allocated to the selected project.	Must	`employees.view` (options need `projects.view`)	REQ-EMP-06
FR-EMP-07	The system shall filter the list by role, offering Employee, Manager and Administrator to everyone and custom roles only to users holding `employees.role.assign`.	Must	`employees.view`, `employees.role.assign`	REQ-EMP-07
FR-EMP-08	The system shall, when custom roles cannot be loaded, list only the standard roles and show "Custom roles could not be loaded. Only the standard roles are listed."	Should	`employees.role.assign`	REQ-EMP-07, REQ-EMP-36
FR-EMP-09	The system shall filter the list by status: All statuses, Active or Inactive.	Must	`employees.view`	REQ-EMP-08
FR-EMP-10	The system shall apply search and all filters together and clear them all when the user clicks **Reset**.	Should	`employees.view`	REQ-EMP-05 – REQ-EMP-09
FR-EMP-11	The system shall show "No employees yet – Add the first employee to get started." when there are no employees, "No employees match these filters – Try a different search or clear the filters." when filters hide every row, and "Could not load employees. Check your connection and try again." when loading fails.	Should	`employees.view`	REQ-EMP-01, REQ-EMP-04

ID	Requirement	Priority	Permission	BRD trace
FR-EMP-12	The system shall show the Employee Management menu item and open `/employees` only for users holding at least one employee action permission, and redirect other users (including Managers holding only `employees.view`) to their dashboard.	Must	Any employee action permission	REQ-EMP-10
FR-EMP-13	The system shall show **Add employee**, **Edit**, **Activate/Deactivate** and **Delete** only to users holding `employees.create`, `employees.edit`, `employees.status.change` and `employees.delete` respectively.	Must	As listed	REQ-EMP-11
FR-EMP-14	The system shall refuse every employee action requested without the matching permission, even when requested outside the screen, and show the action's generic failure message.	Must	As listed	REQ-EMP-11, NFR-EMP-02, BR-04
FR-EMP-15	The system shall never show **Delete** on the signed-in user's own row.	Must	`employees.delete`	REQ-EMP-12, BR-EMP-05
FR-EMP-16	The system shall ask "Deactivate employee?" – "{name} will not be able to sign in until they are activated again." before deactivating, and activate an inactive employee without confirmation.	Must	`employees.status.change`	REQ-EMP-13, NFR-EMP-04
FR-EMP-17	The system shall refuse to deactivate the only active Administrator with "The only active Administrator cannot be deactivated."	Must	System	REQ-EMP-15, BR-EMP-04, BR-26
FR-EMP-18	The system shall, on deactivation, switch the account off, end all of the person's sessions so their next request is rejected, and refuse their sign-in until they are activated again.	Must	`employees.status.change`	REQ-EMP-14, BR-EMP-01, BR-EMP-08, BR-02, NFR-EMP-01
FR-EMP-19	The system shall confirm a status change with "{name} activated" or "{name} deactivated" and refresh the row.	Should	`employees.status.change`	REQ-EMP-13
FR-EMP-20	The system shall ask "Delete employee?" – "{name} will no longer appear in the system or be able to sign in. Their tasks, comments and mail are kept." before deleting.	Must	`employees.delete`	REQ-EMP-16, NFR-EMP-04
FR-EMP-21	The system shall refuse to delete the user's own account with "You cannot delete your own account."	Must	System	REQ-EMP-17, BR-EMP-05, BR-26
FR-EMP-22	The system shall refuse to delete the only active Administrator with "The only active Administrator cannot be deleted."	Must	System	REQ-EMP-19, BR-EMP-04, BR-26
FR-EMP-23	The system shall refuse to delete an employee who still has unfinished assigned tasks with "{name} still has {n} unfinished task(s). Reassign it/them before deleting this employee."	Must	System	REQ-EMP-18, BR-EMP-06

ID	Requirement	Priority	Permission	BRD trace
FR-EMP-24	The system shall delete by soft delete: the person disappears from every list and cannot sign in, while their tasks, comments and mail are kept and stay visible in history and reports.	Must	`employees.delete`	REQ-EMP-16, BR-EMP-07, BR-21
FR-EMP-25	The system shall, on deletion, end all of the person's sessions and remove their project-manager flags and department-manager links.	Must	`employees.delete`	REQ-EMP-20, BR-EMP-08, BR-EMP-14
FR-EMP-26	The system shall keep a deleted person's email and username reserved so that no new or edited employee can use them.	Must	System	REQ-EMP-21, BR-EMP-03
FR-EMP-27	The system shall confirm a deletion with "{name} deleted" and remove the row.	Should	`employees.delete`	REQ-EMP-16

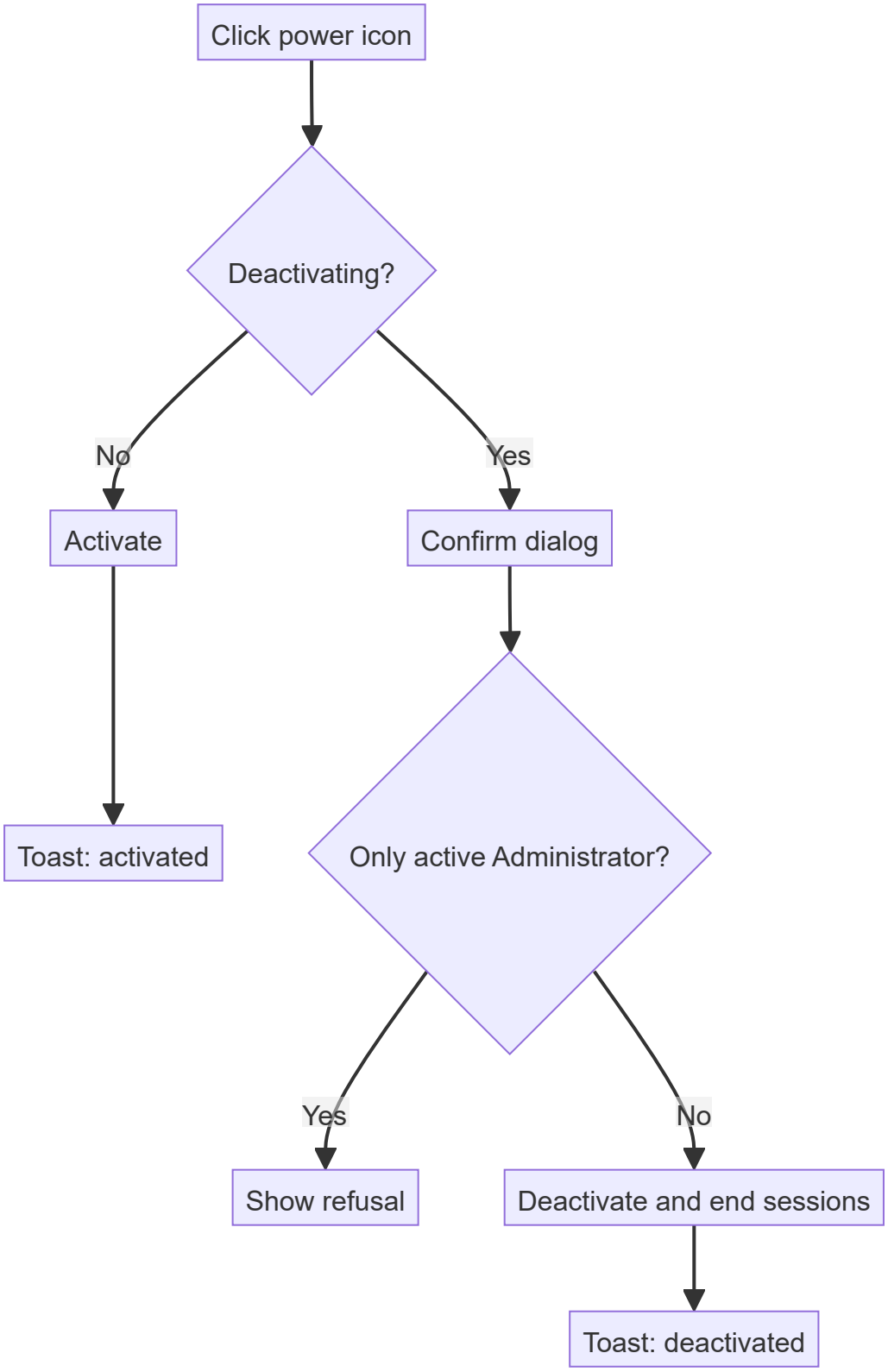
Use case: UC-EMP-01 – Find employees

Item	Detail
Actor	Administrator
Preconditions	Signed in with MFA; holds an employee action permission and `employees.view`
Trigger	Opens **TaskFlow Management** → **Employee Management**
Main flow	1. System lists live employees by full name with the count. 2. Administrator types in **Search** and picks Project, Role or Status. 3. System narrows the list and shows "{n} employees match the current filters". 4. Administrator clicks **Reset**; the full list returns.
Alternate flows	A1: opened from **View employees** on a project card → the Project filter is already set.
Exceptions	E1: no employee matches → "No employees match these filters – Try a different search or clear the filters." E2: load fails → "Could not load employees. Check your connection and try again." E3: user without an action permission opens `/employees` → redirected to their dashboard.
Postconditions	None (read only).

Use case: UC-EMP-02 – Activate or deactivate an employee

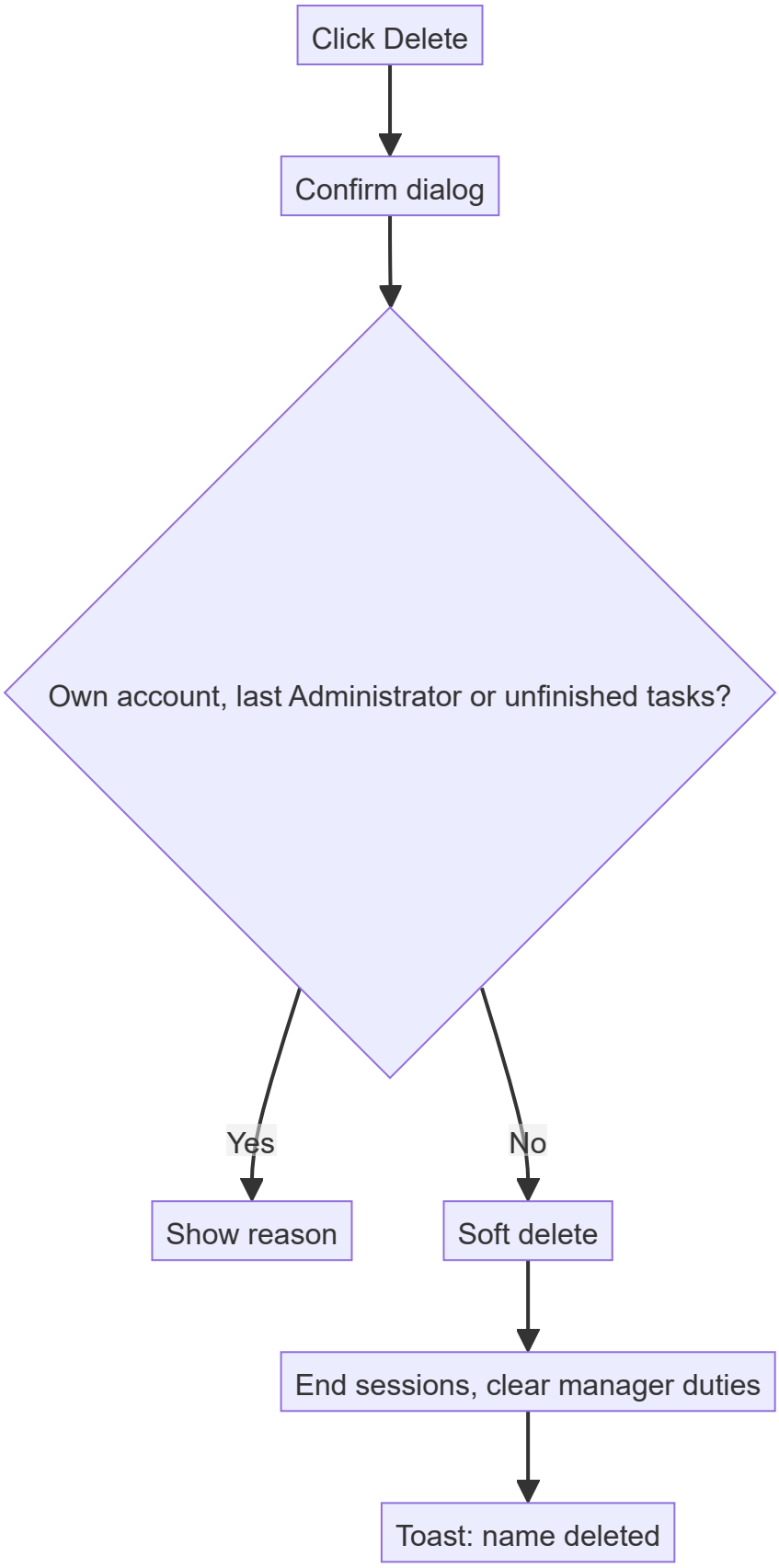
Item	Detail
Actor	Administrator
Preconditions	Holds `employees.status.change`
Trigger	Clicks the power icon on a row
Main flow	1. For an active employee the system asks "Deactivate employee?". 2. Administrator clicks **Deactivate**. 3. System checks the last-Administrator rule, switches the account off and ends all sessions. 4. System shows "{name} deactivated".

Item	Detail
Alternate flows	A1: the employee is inactive → the system activates at once and shows "{name} activated".
Exceptions	E1: only active Administrator → "The only active Administrator cannot be deactivated." E2: other failure → "Could not update the employee status."
Postconditions	Status changed; a deactivated person cannot sign in and their open sessions stop working.



Use case: UC-EMP-03 – Delete an employee

Item	Detail
Actor	Administrator
Preconditions	Holds `employees.delete`; the row is not the user's own
Trigger	Clicks the trash icon on a row
Main flow	1. System asks "Delete employee?". 2. Administrator confirms. 3. System checks own account, last Administrator and unfinished tasks. 4. System soft-deletes the person, ends their sessions and clears their manager duties. 5. System shows "{name} deleted".
Alternate flows	–
Exceptions	E1: own account → "You cannot delete your own account." E2: only active Administrator → "The only active Administrator cannot be deleted." E3: unfinished tasks → "{name} still has {n} unfinished task(s). Reassign it/them before deleting this employee." E4: other failure → "Could not delete the employee."
Postconditions	Person hidden from every list and unable to sign in; history kept; email and username reserved.



Field validations

Field	Rule	Error message
Status change target	Must not be the only active Administrator when deactivating	"The only active Administrator cannot be deactivated."

Field	Rule	Error message
Delete target	Not own account; not the only active Administrator; no unfinished assigned tasks	See FR-EMP-21 – FR-EMP-23

Business rules applied: BR-EMP-01, BR-EMP-03 – BR-EMP-08, BR-EMP-14; BR-02, BR-21, BR-26

Notifications / side effects

Event	Who is notified	Channel
Employee deactivated or deleted	Nobody; the person's sessions end	– (no notification, no email)

Messages

Situation	Message
Activated / deactivated	"{name} activated" / "{name} deactivated"
Deleted	"{name} deleted"
Status change refused	"Could not update the employee status."
Delete refused	"Could not delete the employee."
No permission	Menu item hidden; `/employees` redirects to the dashboard; a refused call shows the action's failure message

4.1.2 Screen: Employee Details (side panel)

Field	Value
Route / location	Side panel on `/employees`, opened by clicking an employee's name
Access	Same as 4.1.1; **Reset MFA** needs `employees.mfa.reset`. Opening another person's single record directly (not from the list) needs `employees.edit`
Entry points	Employee name link in the list

Change from original BRD (Screen 5): the "Employee details view" is kept; derived managers, the Manager's own projects and team, and the MFA reset are new.

Screen elements

Element	Type	Description	Mandatory	Visible to
Employee code, Username, Email, Role	Read-only text	Identity of the person	–	All with screen access
Projects	Read-only list	Project, position and percentage	–	All with screen access
Departments	Read-only list	Department names	–	All with screen access
Managers	Read-only list	Derived from departments and flagged project managers	–	All with screen access
Manages projects / Team	Read-only lists	Shown when the person is a Manager	–	All with screen access

Element	Type	Description	Mandatory	Visible to
Status, Created, Two-step, Last login	Read-only text	Account facts	–	All with screen access
Reset MFA	Button	Removes the person's authenticator	–	Users with `employees.mfa.reset`, only when the person has MFA set up
Close	Button	Closes the panel	–	All with screen access
Edit	Button	Opens Edit employee for this person	–	All with screen access (see section 10)

Functional requirements

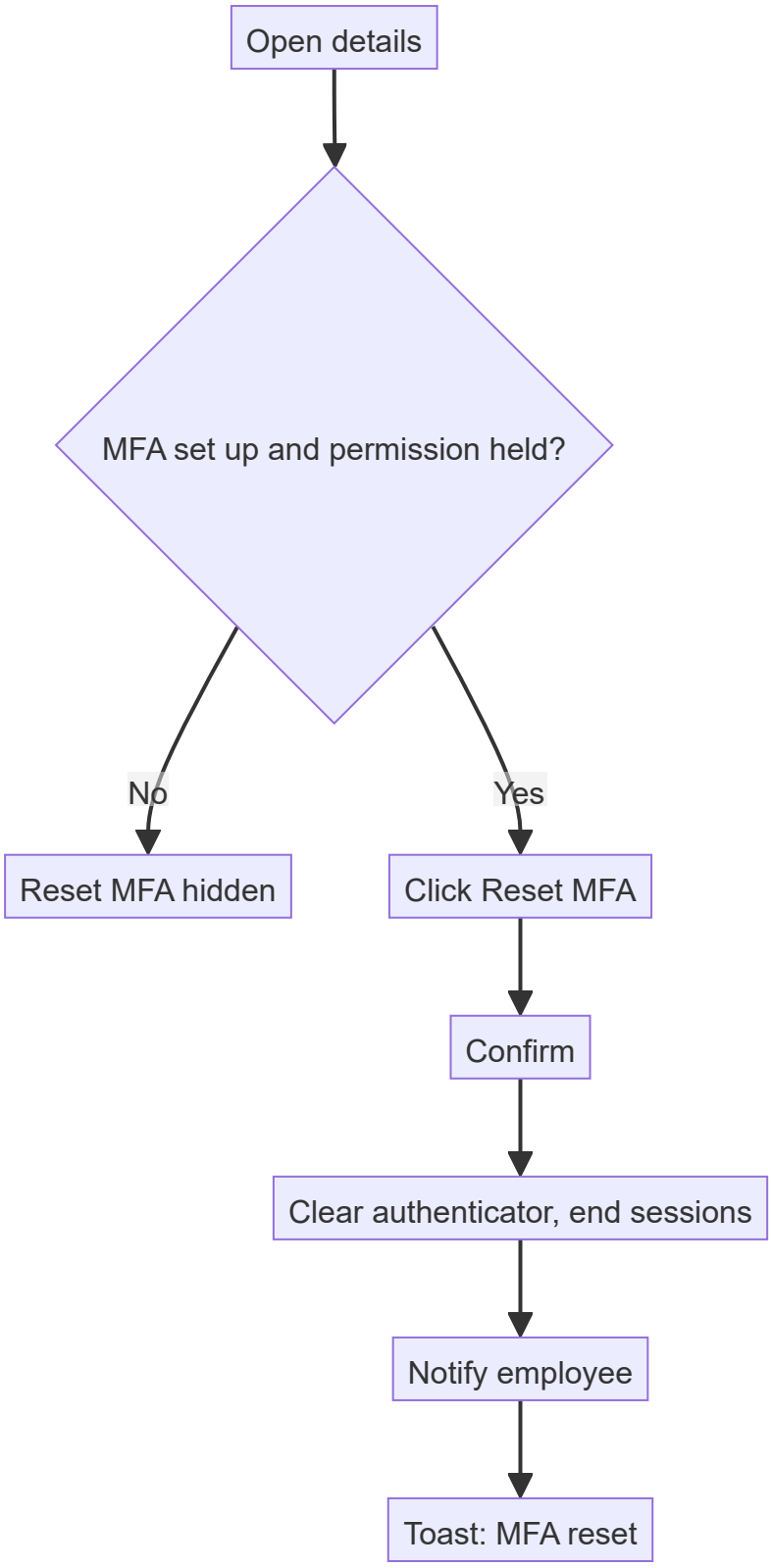
ID	Requirement	Priority	Permission	BRD trace
FR-EMP-28	The system shall open a read-only Employee details side panel when the user clicks an employee's name.	Must	`employees.view`	REQ-EMP-22
FR-EMP-29	The system shall show the employee code, username, email, role, projects (with position and percentage), departments and managers.	Must	`employees.view`	REQ-EMP-23
FR-EMP-30	The system shall show, when the person is a Manager, the projects they manage and their team.	Should	`employees.view`	REQ-EMP-24
FR-EMP-31	The system shall show the account status, created date, two-step verification status and last login.	Should	`employees.view`	REQ-EMP-25
FR-EMP-32	The system shall derive a person's managers as the managers of the departments they belong to plus the flagged project managers of their projects, and shall not accept a typed manager value.	Must	System	REQ-EMP-26, BR-EMP-12
FR-EMP-33	The system shall open the Edit employee side panel for the same person when the user clicks **Edit**.	Must	`employees.edit`	REQ-EMP-22
FR-EMP-34	The system shall show **Reset MFA** only when the person has MFA set up and the user holds `employees.mfa.reset`.	Must	`employees.mfa.reset`	REQ-EMP-27
FR-EMP-35	The system shall ask "Reset two-step verification?" – "{name} will be signed out everywhere and can sign in with just their password. They can set up an authenticator app again from Profile." before resetting.	Must	`employees.mfa.reset`	REQ-EMP-28, NFR-EMP-04
FR-EMP-36	The system shall, after confirmation, remove the person's authenticator, end all of their sessions and let them sign in with their password and set MFA up again.	Must	`employees.mfa.reset`	REQ-EMP-28, BR-EMP-08, BR-25
FR-EMP-37	The system shall send the person the system notification "An administrator reset your authenticator MFA. Sign in with your password and set it up again from Profile."	Must	System	REQ-EMP-29
FR-EMP-38	The system shall confirm with "MFA reset for {Full name}", or show "Could not reset MFA." when the reset is refused.	Should	`employees.mfa.reset`	REQ-EMP-28

Use case: UC-EMP-04 – View an employee's details

Item	Detail
Actor	Administrator
Preconditions	On the Employee Management list
Trigger	Clicks an employee's name
Main flow	1. System opens the details panel. 2. System shows identity, projects, departments and derived managers. 3. For a Manager it also shows managed projects and team. 4. Administrator clicks **Close** or **Edit**.
Alternate flows	A1: **Edit** → Edit employee panel opens (UC-EMP-07).
Exceptions	–
Postconditions	None (read only).

Use case: UC-EMP-05 – Reset an employee's MFA

Item	Detail
Actor	Administrator
Preconditions	Holds `employees.mfa.reset`; the person has MFA set up
Trigger	Clicks **Reset MFA** in the details panel
Main flow	1. System asks "Reset two-step verification?". 2. Administrator confirms. 3. System removes the authenticator and ends all of the person's sessions. 4. System sends the person the MFA-reset notification. 5. System shows "MFA reset for {Full name}".
Alternate flows	A1: the person has no MFA set up → **Reset MFA** is not shown.
Exceptions	E1: refused → "Could not reset MFA."
Postconditions	The person signs in with the password only and sets up a new authenticator from My Profile.



Field validations: *Not applicable for this screen* (read only).

Business rules applied: BR-EMP-08, BR-EMP-12; BR-25

Notifications / side effects

Event	Who is notified	Channel
MFA reset	The employee whose MFA was reset	In-app system notification (N-EMP-01)

Messages

Situation	Message
Success	"MFA reset for {Full name}"
Refused	"Could not reset MFA."

4.1.3 Screen: Add / Edit Employee (side panel)

Field	Value
Route / location	Side panel on `/employees` – titled **Add employee** ("New employee – Fill in the details below to add them to TaskFlow.") or **Edit employee**
Access	Administrator; custom roles with `employees.create` (add) or `employees.edit` (edit); the role field needs `employees.role.assign`; giving or removing the Administrator role needs `roles.administrator.assign` (Administrator role only). Manager and Employee: no access
Entry points	**Add employee** button; **Edit** row icon; **Edit** in the details panel

Change from original BRD (Screen 5): departments are set only when adding; roles other than Employee need the assign-roles permission and the Administrator role is Administrator-only; email uniqueness also covers usernames and deleted people with the fixed `@taskflow.com` domain; project allocations, the temporary password and session end on username or role change are new.

Screen elements

Element	Type	Description	Mandatory	Visible to
Full name	Text input	Person's full name	Yes	Users with `employees.create` / `employees.edit`
Username	Text input	Sign-in name	Yes	Same
Email	Text input + fixed suffix	Part before the fixed `@taskflow.com`	Yes	Same
Employee code	Text input	Optional code	No	Same
Role	Drop-down	Employee by default; locked without `employees.role.assign`	Yes	Same (editable with `employees.role.assign`)
Departments	Tick boxes	Existing departments; view only on edit	No	Same
Project allocation rows	Project drop-down, Position (pick or type), Percentage	One row per project; view only on edit with the hint "Projects are assigned in Project management."	No	Same
Add project	Button	Adds an allocation row (add only)	–	Users with `employees.create`
Allocation bar	Indicator	"X% allocated, Y% remaining"	–	Users with `employees.create`
Temporary password	Password input + strength meter	Add only	Yes (add)	Users with `employees.create`
Status	Drop-down	Active or Inactive (add only)	Yes (add)	Users with `employees.create`

Element	Type	Description	Mandatory	Visible to
Sign-out notice	Text	"Changing the username or role signs this employee out. They must sign in again." (edit only)	–	Users with `employees.edit`
Add employee / Save changes	Button	Submits the panel	–	As above

Functional requirements

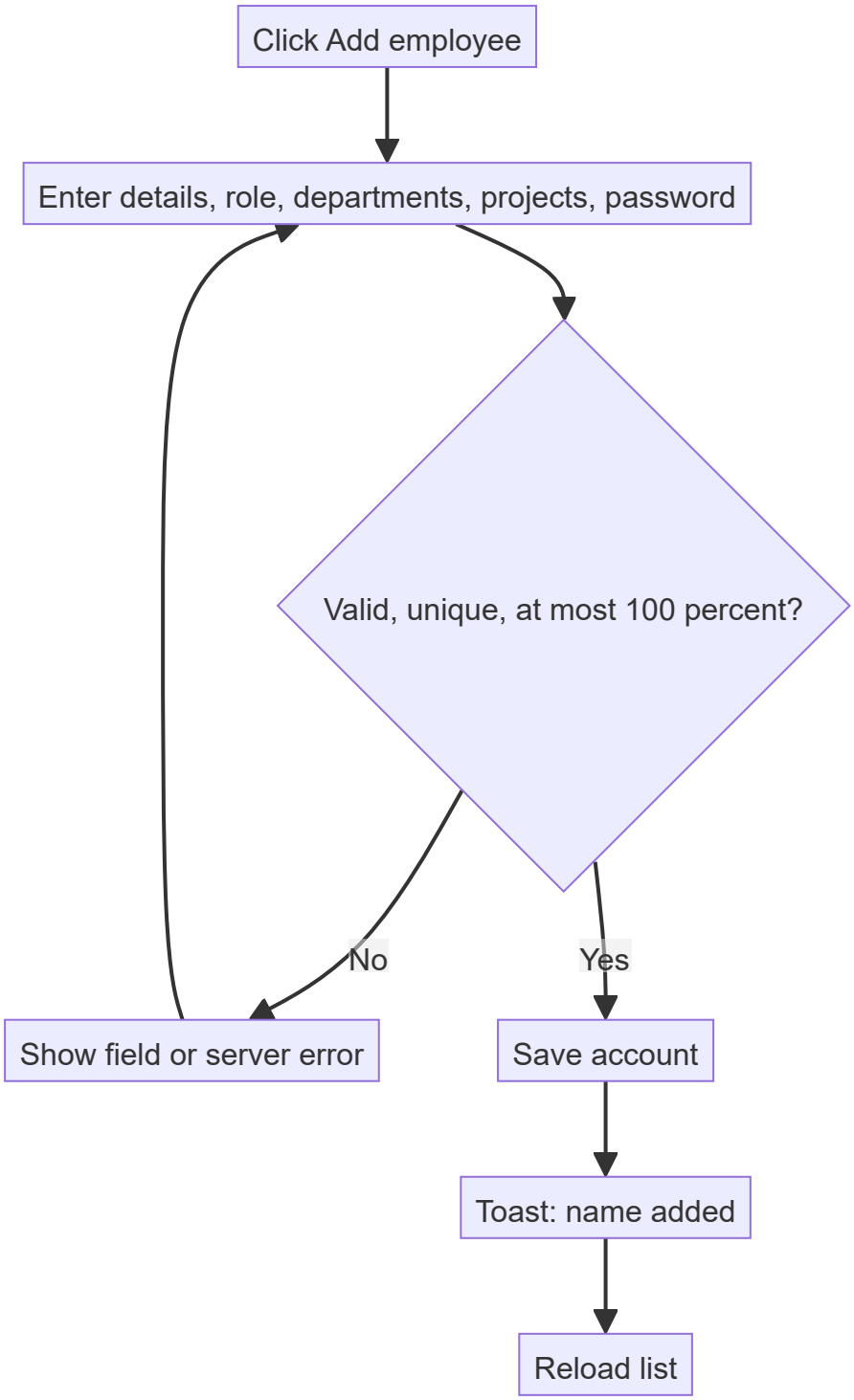
ID	Requirement	Priority	Permission	BRD trace
FR-EMP-39	The system shall create an employee only with a full name (at most 200 characters), a username (at most 100) and an email (at most 256), with an optional employee code.	Must	`employees.create`	REQ-EMP-30
FR-EMP-40	The system shall accept only an email whose part before the fixed `@taskflow.com` holds letters, digits, dot, underscore or hyphen, and store it in lower case.	Must	`employees.create`, `employees.edit`	REQ-EMP-31, BR-EMP-02, BR-24
FR-EMP-41	The system shall reject an email whose part before `@` is already used by any employee, including deleted ones, with "Email already exists."	Must	System	REQ-EMP-32, BR-EMP-03
FR-EMP-42	The system shall reject a username already used by any employee, ignoring case and including deleted ones, with "Email or username already exists."	Must	System	REQ-EMP-33, BR-EMP-03
FR-EMP-43	The system shall keep email and username unique even when two users save at the same moment.	Must	System	REQ-EMP-32, REQ-EMP-33, NFR-EMP-06
FR-EMP-44	The system shall require a temporary password of at least 6 characters when adding, show its strength while typing, and store it only hashed.	Must	`employees.create`	REQ-EMP-34, NFR-EMP-03
FR-EMP-45	The system shall let the user choose the initial status, Active or Inactive, when adding.	Should	`employees.create`	REQ-EMP-35
FR-EMP-46	The system shall give a new employee the Employee role unless the user holds `employees.role.assign` and picks another role; without it the role field is locked and a role change is refused with "Changing a role needs the assign roles permission."	Must	`employees.role.assign`	REQ-EMP-36, BR-EMP-09
FR-EMP-47	The system shall allow giving or removing the Administrator role only to users holding `roles.administrator.assign`.	Must	`roles.administrator.assign`	REQ-EMP-37, BR-EMP-09
FR-EMP-48	The system shall let the user tick existing departments when adding, and reject an unknown department with "Department does not exist."	Must	`employees.create` (list needs `departments.view`)	REQ-EMP-38, BR-EMP-11

ID	Requirement	Priority	Permission	BRD trace
FR-EMP-49	The system shall let the user add allocation rows when adding, each with an existing project, a position picked or newly typed, and a whole percentage from 1 to 100.	Must	`employees.create` (lists need `projects.view`)	REQ-EMP-39, BR-EMP-10
FR-EMP-50	The system shall reject an unknown role, project or position with "Role does not exist.", "Project does not exist." or "Position does not exist."	Must	System	REQ-EMP-36, REQ-EMP-39
FR-EMP-51	The system shall allow a project only once per employee.	Must	System	REQ-EMP-40, BR-EMP-10
FR-EMP-52	The system shall show "X% allocated, Y% remaining" while allocations are edited and reject a total above 100%.	Must	System	REQ-EMP-41, BR-EMP-10
FR-EMP-53	The system shall, when editing, show departments and project allocations view only with the hint "Projects are assigned in Project management." and change only the code, full name, username, email and role.	Must	`employees.edit`	REQ-EMP-42, BR-EMP-11
FR-EMP-54	The system shall never change a person's password from the edit panel, even if a password is sent.	Must	System	REQ-EMP-43, BR-EMP-13
FR-EMP-55	The system shall show the notice "Changing the username or role signs this employee out. They must sign in again." on the edit panel.	Should	`employees.edit`	REQ-EMP-44
FR-EMP-56	The system shall end all of a person's sessions when their username or role changes and confirm with "{Full name} updated. They must sign in again."	Must	`employees.edit`	REQ-EMP-45, BR-EMP-08, NFR-EMP-01
FR-EMP-57	The system shall sign an administrator out at once and take them to the login page when they change their own username or role.	Must	System	REQ-EMP-46
FR-EMP-58	The system shall refuse to move the only active Administrator to another role with "The only active Administrator cannot be moved to another role."	Must	System	REQ-EMP-47, BR-EMP-04, BR-26
FR-EMP-59	The system shall remove a person's project-manager flags and department-manager links when their role changes away from Manager.	Must	System	REQ-EMP-48, BR-EMP-14
FR-EMP-60	The system shall confirm a save with "{Full name} added" or "{Full name} updated", close the panel and reload the list; a refused save shows "Could not add the employee." or "Could not update the employee."	Should	`employees.create`, `employees.edit`	REQ-EMP-49

Use case: UC-EMP-06 – Add an employee

Item	Detail
Actor	Administrator
Preconditions	Holds `employees.create` (and `projects.view`, `departments.view` for the lists)

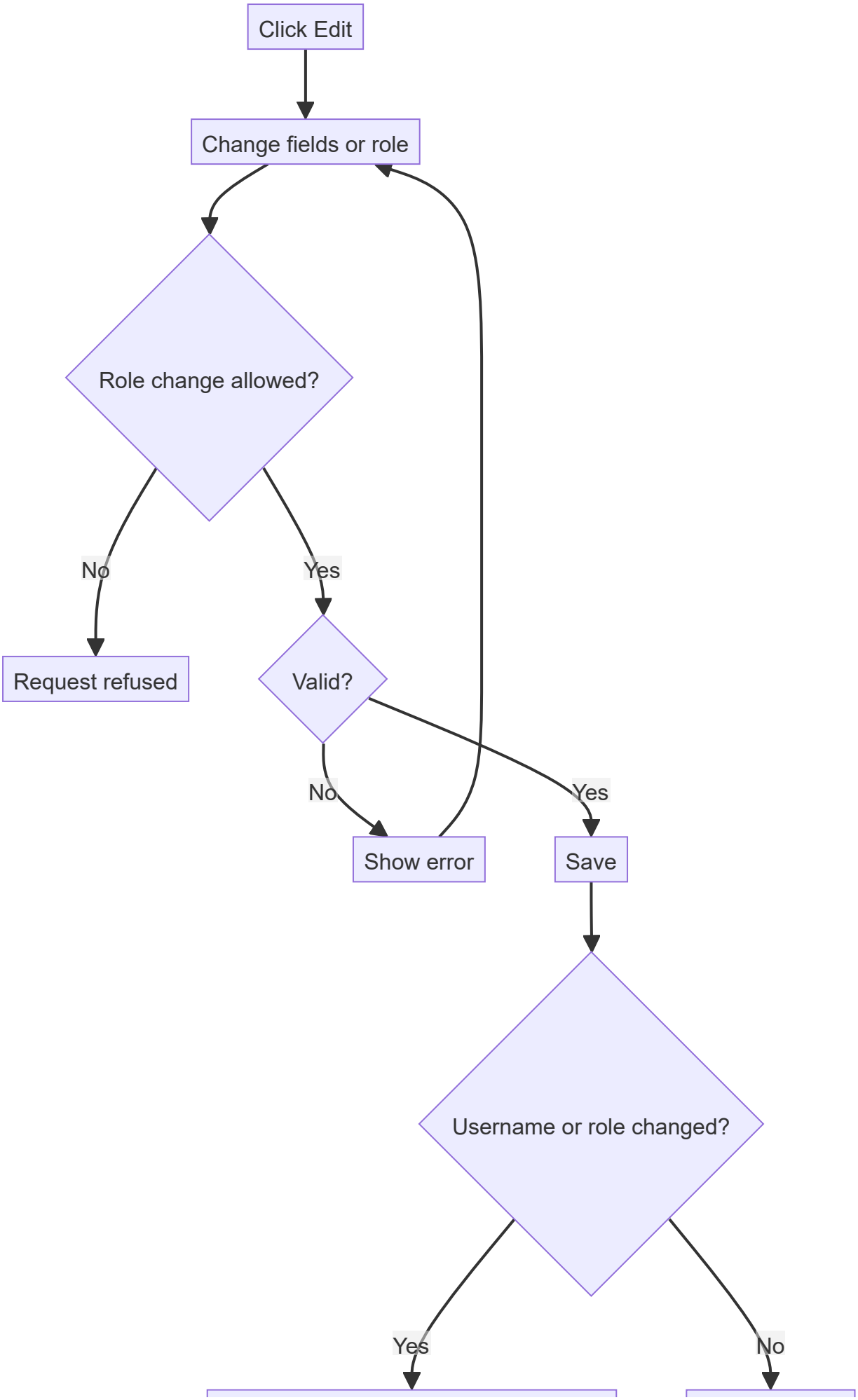
Item	Detail
Trigger	Clicks **Add employee**
Main flow	1. System opens the Add employee panel. 2. Administrator enters Full name, Username, Email (before `@taskflow.com`) and optional Employee code. 3. Administrator picks a Role (if allowed), ticks Departments and adds project rows with position and percentage; the bar shows "X% allocated, Y% remaining". 4. Administrator enters a Temporary password and Status and clicks **Add employee**. 5. System validates and saves. 6. System shows "{Full name} added" and reloads the list.
Alternate flows	A1: no `employees.role.assign` → role stays Employee. A2: a typed position does not exist → it is created and used (see Project Management FRD).
Exceptions	E1: required field empty → field message. E2: duplicate email → "Email already exists." E3: duplicate username → "Email or username already exists." E4: total above 100% → "Allocations total {n}%. A person can be allocated at most 100%." E5: refused → "Could not add the employee."
Post conditions	A sign-in account exists with role, departments and allocations; the temporary password is given to the person outside TaskFlow.



Use case: UC-EMP-07 – Edit an employee (including a role change)

Item	Detail
Actor	Administrator
Preconditions	Holds `employees.edit` (role change: `employees.role.assign`; Administrator role: `roles.administrator.assign`)
Trigger	Clicks **Edit** on a row or in the details panel

Item	Detail
Main flow	1. System opens Edit employee with saved data; departments and allocations view only. 2. Administrator changes code, name, username, email or role. 3. Administrator clicks **Save changes**. 4. System validates, checks role rights and the last-Administrator rule, and saves. 5. If username or role changed, system ends the person's sessions and shows "{Full name} updated. They must sign in again."; otherwise "{Full name} updated".
Alternate flows	A1: own username or role changed → the administrator is signed out and taken to the login page. A2: role changed away from Manager → project-manager flags and department-manager links removed.
Exceptions	E1: only active Administrator moved → "The only active Administrator cannot be moved to another role." E2: role change without permission → "Changing a role needs the assign roles permission." E3: refused → "Could not update the employee."
Post conditions	Details saved; access changes take effect on the person's next request.


Click Edit
Change fields or role
Role change allowed?
No
Yes
Request refused
Valid?
No
Yes
Show error
Save
Username or role changed?
Yes
No

End sessions, toast: must sign in again

Toast: updated

Field validations

Field	Rule	Error message
Full name	Required; at most 200 characters	"Full name is required."
Username	Required; at most 100 characters; unique ignoring case, including deleted people	"Username is required." · "Email or username already exists."
Email	Required; at most 256 characters; letters, digits, dot, underscore or hyphen before `@taskflow.com`; unique by the part before `@`, including deleted people	"Email is required." · "Use letters, digits, dot, underscore or hyphen before @taskflow.com." (form) / "Email must be a name using letters, digits, dot, underscore or hyphen followed by @taskflow.com." (server) · "Email already exists."
Temporary password	Required on add; at least 6 characters	"Password is required." · "Use at least 6 characters."
Role	Must exist; non-Employee needs `employees.role.assign`; Administrator needs `roles.administrator.assign`; not moving the only active Administrator	"Role does not exist." · "Changing a role needs the assign roles permission." · "The only active Administrator cannot be moved to another role."
Departments	Each must exist	"Department does not exist."
Allocation row	Project, position and percentage all set	"Give every project row a project, a position and a whole percentage from 1 to 100."
Allocation percentage	Whole number 1–100	"Allocation percentage must be a whole number from 1 to 100."
Allocation project	Once per person; must exist	"A project can only appear once." (form) / "A project can only appear once for a person." (server) · "Project does not exist."
Allocation position	Must exist (or is created when typed)	"Position does not exist."
Allocation total	At most 100%	"Allocations total {n}%. A person can be allocated at most 100%." (form) / "Allocations total {n}%, which is over 100%. A person can be allocated at most 100% in total." (server)

Business rules applied: BR-EMP-02, BR-EMP-03, BR-EMP-04, BR-EMP-08 – BR-EMP-11, BR-EMP-13, BR-EMP-14; BR-24, BR-26, BR-28

Notifications / side effects

Event	Who is notified	Channel
Employee added or updated	Nobody	– (no notification, no email)
Username or role changed	The person is signed out	Session end only

Messages

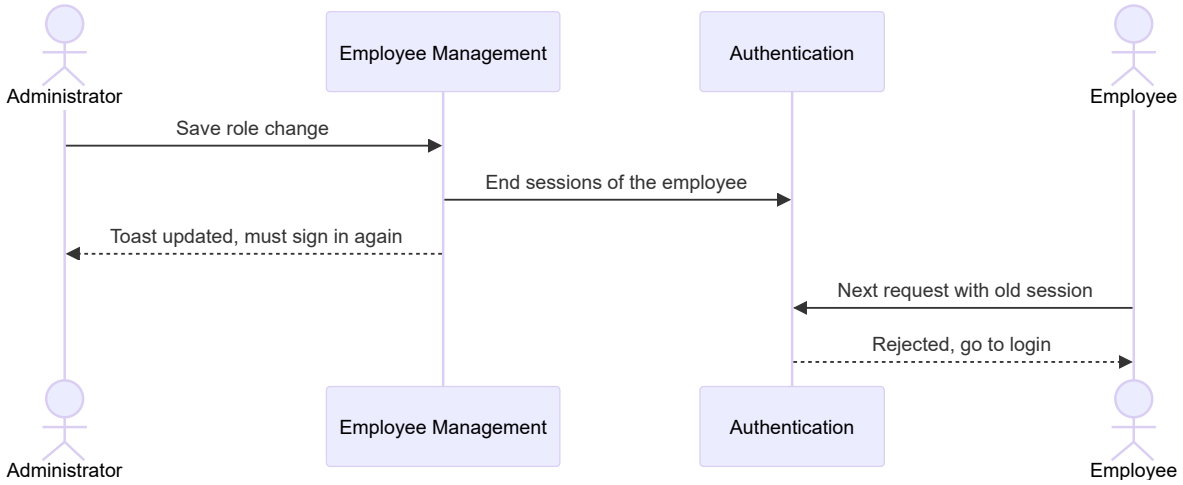
Situation	Message
Success	"{Full name} added" · "{Full name} updated" · "{Full name} updated. They must sign in again."
Roles list failed	"Custom roles could not be loaded. Only the standard roles are listed."
Refused	"Could not add the employee." / "Could not update the employee."
No permission	Buttons hidden; a refused call shows the failure message above

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5. Cross-Module Workflows

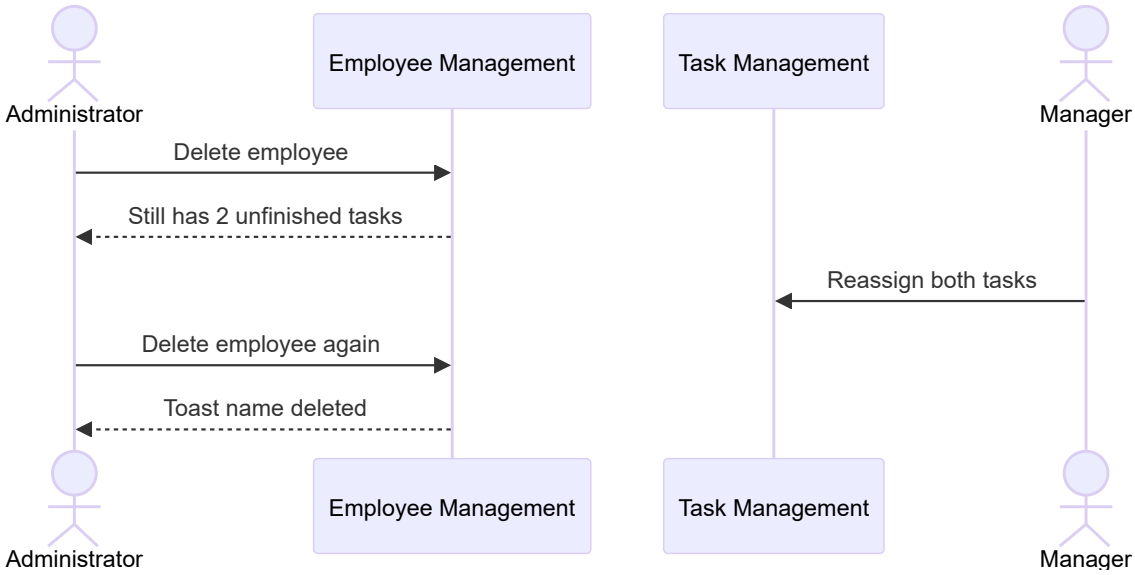
5.1 Role change takes effect at once

Step	Module	Actor / System	Action	FR reference
1	EMP	Administrator	Changes Ravi Menon's role from Employee to Manager and saves	FR-EMP-46, FR-EMP-56
2	EMP	System	Ends all of Ravi's sessions	FR-EMP-56
3	AUTH	System	Rejects Ravi's next request and sends him to the login page	Overall FRD (Authentication)
4	AUTH	Ravi	Signs in again and receives the Manager permissions	Overall FRD (Authentication)



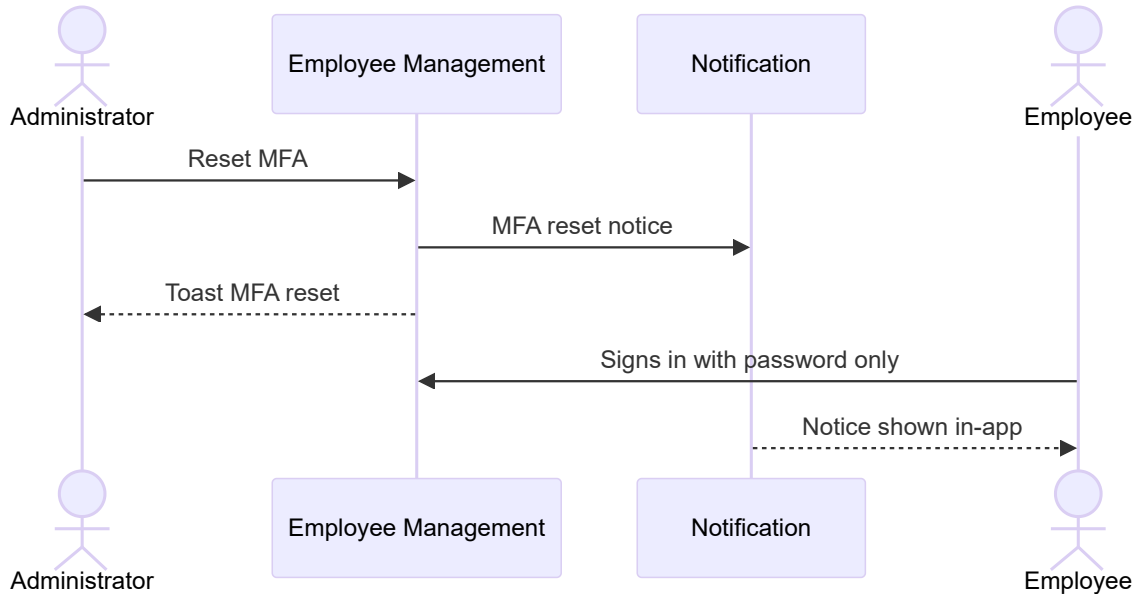
5.2 A person leaves the organisation

Step	Module	Actor / System	Action	FR reference
1	EMP	Administrator	Tries to delete Arun, who has 2 tasks In Progress	FR-EMP-23
2	TASK	Manager	Reassigns both tasks with a reason	Task Management FRD
3	EMP	Administrator	Deletes Arun	FR-EMP-24, FR-EMP-25
4	RPT	System	Keeps showing Arun's past tasks and comments	FR-EMP-24; Reports FRD



5.3 Lost authenticator

Step	Module	Actor / System	Action	FR reference
1	EMP	Administrator	Resets Sanjay's MFA	FR-EMP-35, FR-EMP-36
2	NOTIF	System	Delivers the MFA-reset notification	FR-EMP-37; Notification FRD
3	AUTH / PROF	Sanjay	Signs in with his password and sets up a new authenticator from My Profile	Overall FRD; My Profile FRD



6. Reports & Exports

Not applicable for this sub-module. The list header count is the only summary; Reports keep showing deleted people's history (see the Reports FRD).

7. Notifications & Messages Catalogue

Code	Trigger	Recipients	Channel	Text
N-EMP-01	Administrator resets a person's MFA	The person	In-app system notification	"An administrator reset your authenticator MFA. Sign in with your password and set it up again from Profile."
M-EMP-01	Employee added	Acting user	Toast	"{Full name} added"
M-EMP-02	Employee updated	Acting user	Toast	"{Full name} updated" / "{Full name} updated. They must sign in again."
M-EMP-03	Status changed	Acting user	Toast	"{name} activated" / "{name} deactivated"
M-EMP-04	Employee deleted	Acting user	Toast	"{name} deleted"
M-EMP-05	MFA reset	Acting user	Toast	"MFA reset for {Full name}"

No other employee action sends a notification or an email.

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8. Data Requirements (functional view)

Data item	Description	Created by	Rules
Employee account	Code, full name, username, email, role, active status, created / last-login dates, two-step status	Administrator (add)	Soft delete; email (by part before `@`) and username (ignoring case) unique for ever
Email	Sign-in and contact address	Administrator	Fixed `@taskflow.com` domain; stored in lower case
Temporary password	First password	Administrator (add only)	At least 6 characters; stored hashed; never shown or changed on edit
Project allocation	Project, position, percentage, project-manager flag	Administrator (add); then Project Management	Whole 1–100; project once per person; total at most 100%
Department membership	Departments the person belongs to	Administrator (add); then Department Management	Departments must exist
Derived managers	Department managers plus flagged project managers	System	Never typed in

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9. Non-Functional Requirements (functional impact)

The global NFRs of the overall BRD apply (notably NFR-04 passwords, NFR-08 server validation, NFR-12 usability).

ID	Requirement	Affects
NFR-EMP-01	After a deactivation, deletion, username or role change, or MFA reset, the person's next request with an old session is rejected	FR-EMP-18, FR-EMP-25, FR-EMP-36, FR-EMP-56
NFR-EMP-02	Permission checks are enforced by the server, not only by hidden buttons	FR-EMP-14
NFR-EMP-03	Temporary passwords are stored hashed and never shown	FR-EMP-44
NFR-EMP-04	Deactivate, Delete and Reset MFA show a confirmation dialog	FR-EMP-16, FR-EMP-20, FR-EMP-35
NFR-EMP-05	Every refusal states the reason in plain words	Messages in section 4
NFR-EMP-06	No two accounts share an email local part or username, even under simultaneous saves	FR-EMP-43

10. Assumptions & Open Issues

Type	Description	Owner
Assumption	One organisation with the fixed email domain `@taskflow.com`.	Jojo George
Assumption	The administrator gives the temporary password to the new employee outside TaskFlow.	Jojo George
Open issue	⚠️ To be confirmed: the details panel always shows **Edit**, even to users without `employees.edit` (the save is refused by the server). Should the button be hidden? (R-EMP-06)	Jojo George
Open issue	⚠️ To be confirmed: the TaskFlow Assistant's role-change path may not require `employees.role.assign` or `roles.administrator.assign`. Is that intended? (R-EMP-05)	Jojo George
Open issue	⚠️ To be confirmed: the role-permission matrix lists `roles.administrator.assign` as configurable for custom roles, while the permission catalogue marks it not assignable. This FRD follows the catalogue.	Jojo George
Open issue	⚠️ To be confirmed: the catalogue describes `employees.delete` as "Permanently remove an employee account", but deletion is a soft delete (FR-EMP-24).	Jojo George
Open issue	The search placeholder says "Name or email" but the search also matches username (FR-EMP-05).	Jojo George
Open issue	A custom role with employee rights but without `projects.view` / `departments.view` sees empty Project filter and form lists (R-EMP-04).	Jojo George
Open issue	Opening another person's single record directly needs `employees.edit`, while the list shows the same data with `employees.view`.	Jojo George

11. Requirements Traceability Matrix

Test evidence lists the automated test suites that cover the behaviour; the mapping was checked against the test sources. Frontend specs (`*.spec.ts`) are under `frontend/src/app`. "Manual UI" = screen behaviour checked by hand.

BRD item	FRD requirement(s)	Screen	Use case	Test case
REQ-EMP-01	FR-EMP-01, FR-EMP-11	4.1.1	UC-EMP-01	UsersEndpointsTests, AdminDeleteEndpointsTests, employee-list.component.spec.ts
REQ-EMP-02	FR-EMP-02	4.1.1	UC-EMP-01	employee-list.component.spec.ts
REQ-EMP-03	FR-EMP-03	4.1.1	UC-EMP-01	employee-list.component.spec.ts
REQ-EMP-04	FR-EMP-04, FR-EMP-11	4.1.1	UC-EMP-01	employee-list.component.spec.ts
REQ-EMP-05	FR-EMP-05, FR-EMP-10	4.1.1	UC-EMP-01	employee-list.component.spec.ts (server-side matching not covered)
REQ-EMP-06	FR-EMP-06, FR-EMP-10	4.1.1	UC-EMP-01	UserAllocationServiceTests, ProjectTeamEndpointsTests, employee-list.component.spec.ts, allocation-services.spec.ts
REQ-EMP-07	FR-EMP-07, FR-EMP-08, FR-EMP-10	4.1.1	UC-EMP-01	RoleAssignmentSafeguardTests (role options); Manual UI (filter and fallback message)
REQ-EMP-08	FR-EMP-09, FR-EMP-10	4.1.1	UC-EMP-01	employee-list.component.spec.ts
REQ-EMP-09	FR-EMP-10	4.1.1	UC-EMP-01	Manual UI
REQ-EMP-10	FR-EMP-12	4.1.1	UC-EMP-01	app.routes.spec.ts, nav-items.spec.ts
REQ-EMP-11	FR-EMP-13, FR-EMP-14	4.1.1	UC-EMP-02, UC-EMP-03	UsersEndpointsTests, AdminDeleteEndpointsTests, PermissionAuthorizationTests, employee-list.component.spec.ts
REQ-EMP-12	FR-EMP-15	4.1.1	UC-EMP-03	employee-list.component.spec.ts
REQ-EMP-13	FR-EMP-16, FR-EMP-19	4.1.1	UC-EMP-02	employee-list.component.spec.ts
REQ-EMP-14	FR-EMP-18	4.1.1	UC-EMP-02	UsersEndpointsTests

BRD item	FRD requirement(s)	Screen	Use case	Test case
REQ-EMP-15	FR-EMP-17	4.1.1	UC-EMP-02	LastAdministratorStatusTests
REQ-EMP-16	FR-EMP-20, FR-EMP-24, FR-EMP-27	4.1.1	UC-EMP-03	UserDeletionTests, AdminDeleteEndpointsTests, employee-list.component.spec.ts
REQ-EMP-17	FR-EMP-21	4.1.1	UC-EMP-03	UserDeletionTests, AdminDeleteEndpointsTests
REQ-EMP-18	FR-EMP-23	4.1.1	UC-EMP-03	UserDeletionTests, AdminDeleteEndpointsTests
REQ-EMP-19	FR-EMP-22	4.1.1	UC-EMP-03	UserDeletionTests
REQ-EMP-20	FR-EMP-25	4.1.1	UC-EMP-03	UserDeletionTests, AdminDeleteEndpointsTests
REQ-EMP-21	FR-EMP-26	4.1.1	UC-EMP-03	UserDeletionTests, EmailDomainTests
REQ-EMP-22	FR-EMP-28, FR-EMP-33	4.1.2	UC-EMP-04	employee-list.component.spec.ts, employee-form-panel.component.spec.ts
REQ-EMP-23	FR-EMP-29	4.1.2	UC-EMP-04	employee-form-panel.component.spec.ts
REQ-EMP-24	FR-EMP-30	4.1.2	UC-EMP-04	ManagerAssignmentTests, employee-form-panel.component.spec.ts
REQ-EMP-25	FR-EMP-31	4.1.2	UC-EMP-04	employee-mfa-reset.spec.ts (two-step status); Manual UI (other fields)
REQ-EMP-26	FR-EMP-32	4.1.2	UC-EMP-04	ManagerAssignmentTests, ManagerAssignmentEndpointsTests, employee-form-panel.component.spec.ts
REQ-EMP-27	FR-EMP-34	4.1.2	UC-EMP-05	employee-mfa-reset.spec.ts
REQ-EMP-28	FR-EMP-35, FR-EMP-36, FR-EMP-38	4.1.2	UC-EMP-05	MfaAdminResetEndpointsTests, MfaServiceTests, employee-mfa-reset.spec.ts
REQ-EMP-29	FR-EMP-37	4.1.2	UC-EMP-05	MfaAdminResetEndpointsTests, MfaServiceTests

BRD item	FRD requirement(s)	Screen	Use case	Test case
REQ-EMP-30	FR-EMP-39	4.1.3	UC-EMP-06	UsersEndpointsTests, employee-form-panel.component.spec.ts
REQ-EMP-31	FR-EMP-40	4.1.3	UC-EMP-06, UC-EMP-07	EmailDomainTests, employee-form-panel.component.spec.ts
REQ-EMP-32	FR-EMP-41, FR-EMP-43	4.1.3	UC-EMP-06	EmailDomainTests, employee-form-panel.component.spec.ts (409 message)
REQ-EMP-33	FR-EMP-42, FR-EMP-43	4.1.3	UC-EMP-06	EmailDomainTests, employee-form-panel.component.spec.ts (409 message)
REQ-EMP-34	FR-EMP-44	4.1.3	UC-EMP-06	UsersEndpointsTests, employee-form-panel.component.spec.ts
REQ-EMP-35	FR-EMP-45	4.1.3	UC-EMP-06	employee-form-panel.component.spec.ts
REQ-EMP-36	FR-EMP-08, FR-EMP-46, FR-EMP-50	4.1.3	UC-EMP-06, UC-EMP-07	UsersEndpointsTests, RoleAssignmentSafeguardTests, UserAllocationServiceTests, employee-form-panel.component.spec.ts
REQ-EMP-37	FR-EMP-47	4.1.3	UC-EMP-07	RoleAssignmentSafeguardTests
REQ-EMP-38	FR-EMP-48	4.1.3	UC-EMP-06	UserAllocationServiceTests, employee-form-panel.component.spec.ts
REQ-EMP-39	FR-EMP-49, FR-EMP-50	4.1.3	UC-EMP-06	UserAllocationServiceTests, employee-form-panel.component.spec.ts
REQ-EMP-40	FR-EMP-51	4.1.3	UC-EMP-06	UserAllocationServiceTests, employee-form-panel.component.spec.ts
REQ-EMP-41	FR-EMP-52	4.1.3	UC-EMP-06	UserAllocationServiceTests, employee-form-panel.component.spec.ts
REQ-EMP-42	FR-EMP-53	4.1.3	UC-EMP-07	employee-form-panel.component.spec.ts
REQ-EMP-43	FR-EMP-54	4.1.3	UC-EMP-07	UsersEndpointsTests, RoleAssignmentSafeguardTests, employee-form-panel.component.spec.ts
REQ-EMP-44	FR-EMP-55	4.1.3	UC-EMP-07	employee-form-panel.component.spec.ts

BRD item	FRD requirement(s)	Screen	Use case	Test case
REQ-EMP-45	FR-EMP-56	4.1.3	UC-EMP-07	UsersEndpointsTests
REQ-EMP-46	FR-EMP-57	4.1.3	UC-EMP-07	employee-form-panel.component.spec.ts
REQ-EMP-47	FR-EMP-58	4.1.3	UC-EMP-07	LastAdministratorTests
REQ-EMP-48	FR-EMP-59	4.1.3	UC-EMP-07	ManagerAssignmentTests
REQ-EMP-49	FR-EMP-60	4.1.3	UC-EMP-06, UC-EMP-07	employee-form-panel.component.spec.ts
BR-EMP-01	FR-EMP-18	4.1.1	UC-EMP-02	UsersEndpointsTests
BR-EMP-02	FR-EMP-40	4.1.3	UC-EMP-06	EmailDomainTests
BR-EMP-03	FR-EMP-26, FR-EMP-41, FR-EMP-42	4.1.1, 4.1.3	UC-EMP-03, UC-EMP-06	EmailDomainTests, UserDeletionTests
BR-EMP-04	FR-EMP-17, FR-EMP-22, FR-EMP-58	4.1.1, 4.1.3	UC-EMP-02, UC-EMP-03, UC-EMP-07	LastAdministratorTests, LastAdministratorStatusTests, UserDeletionTests
BR-EMP-05	FR-EMP-15, FR-EMP-21	4.1.1	UC-EMP-03	AdminDeleteEndpointsTests, UserDeletionTests, employee-list.component.spec.ts
BR-EMP-06	FR-EMP-23	4.1.1	UC-EMP-03	UserDeletionTests
BR-EMP-07	FR-EMP-01, FR-EMP-24	4.1.1	UC-EMP-03	UserDeletionTests, AdminDeleteEndpointsTests
BR-EMP-08	FR-EMP-18, FR-EMP-25, FR-EMP-36, FR-EMP-56	4.1.1 – 4.1.3	UC-EMP-02, UC-EMP-03, UC-EMP-05, UC-EMP-07	UsersEndpointsTests, UserDeletionTests, MfaAdminResetEndpointsTests
BR-EMP-09	FR-EMP-46, FR-EMP-47	4.1.3	UC-EMP-06, UC-EMP-07	UsersEndpointsTests, RoleAssignmentSafeguardTests
BR-EMP-10	FR-EMP-49, FR-EMP-51, FR-EMP-52	4.1.3	UC-EMP-06	UserAllocationServiceTests

BRD item	FRD requirement(s)	Screen	Use case	Test case
BR-EMP-11	FR-EMP-48, FR-EMP-53	4.1.3	UC-EMP-06, UC-EMP-07	UserAllocationServiceTests, employee-form-panel.component.spec.ts
BR-EMP-12	FR-EMP-32	4.1.2	UC-EMP-04	ManagerAssignmentTests
BR-EMP-13	FR-EMP-54	4.1.3	UC-EMP-07	UsersEndpointsTests, RoleAssignmentSafeguardTests, employee-form-panel.component.spec.ts
BR-EMP-14	FR-EMP-25, FR-EMP-59	4.1.1, 4.1.3	UC-EMP-03, UC-EMP-07	ManagerAssignmentTests, UserDeletionTests

Coverage: 49 of 49 `REQ-EMP` requirements and 14 of 14 `BR-EMP` rules trace to at least one FR; 0 untraced.

12. Approvals

Role	Name	Decision	Date
Product owner	Jojo George	Pending	–
Reviewer	Jojo George	Pending	–

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